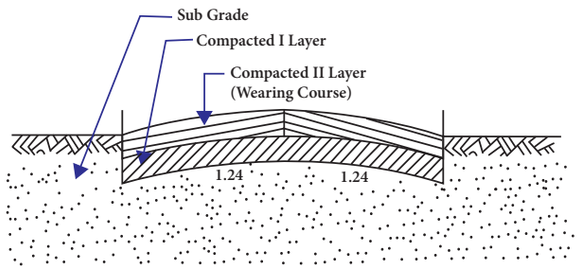
6.5 Highway Construction

6.5.1 Types of Highway Construction

1. Earthern Road
2. Gravel Road
3. Water Bound Macadam Road
4. Bituminous Road
5. Cement Concrete Road

1. Construction of Earthern Road

It is the cheapest type of road in construction because it is made from the natural soil available at the site. It provides a link for very light traffic.



CROSS SECTION OF EARTH ROAD

Construction

1. The centre line is fixed and reference pegs are driven for the guidance of vertical profile of road.
2. The ground is cleared by removing all vegetation.
3. The subgrade is prepared and it is provided with necessary camber (1 in 24) and longitudinal gradient.
4. The subgrade is properly compacted by rolling.
5. A layer of 10cm graded soil is spread evenly and rolled at optimum moisture content with sheep foot rollers.
6. If required, another soil layer of 10cm thick is also spread and rolled properly to act as wearing course.
7. The surface is water cured for 4 or 5 days without traffic.
8. The compacted earthern road is allowed to dry for a period of about 5 to 10 days before opening it to traffic.



Maintenance of Earthern Road

Proper and constant maintenance is required to keep these roads in service. Hence, periodic repairs to pot holes and ruts are essential which are filled with earth and compacted by hand rammers. Side drains are also properly maintained as the life and efficiency of the earthern roads mainly depend on the efficient drainage system.

2. Construction of Gravel Road

Gravel roads are considered superior to earthern roads as they can carry heavier traffic. These roads are constructed with well graded quality gravel obtainable from river beds or by crushing the stones. The various methods of construction of gravel road are

i. Trench spread method

ii. Feather edge spread method

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